

BEFORE WE START

You are going to need a book and pen for your learning journey. Why not buy these?
(Affiliate links mama ivi, niku em cost padadu but miru konte naku commission osthadi 5rs lol)

Notebook - <https://amzn.to/4tYSyYi>

Ball Pen - <https://amzn.to/4tKNGWn>

BASIC SHELL COMMANDS

Command line → bash

pwd → pathname of current working directory

ls → list

ls -l → long list

ls -la → list hidden files too

cd → change path

~ → home directory

cd .. → oka directory venakki vellu

FILE AND DIRECTORY OPERATIONS

`mkdir xxx(name)` → make a directory

`mkdir -p xxx/yyy/zzz` → make nested directories

`touch xxx` → create empty file

`cp source dest` → copy stuff

`cp -r dir1 dir2` → recursive copy

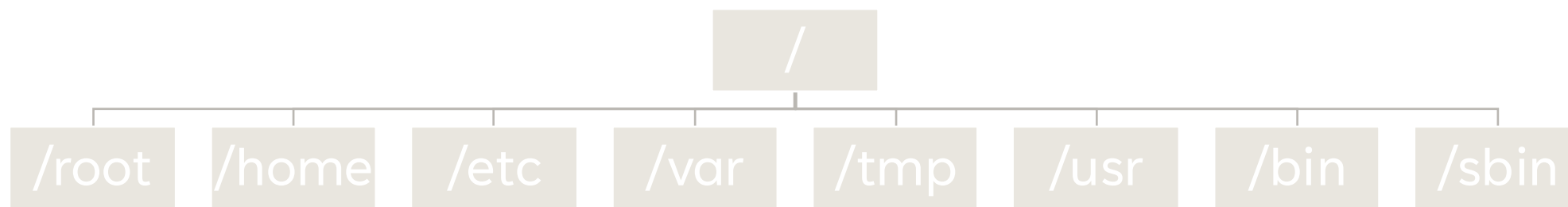
`mv oldfile newfile` → move or rename

`rm file` → remove file

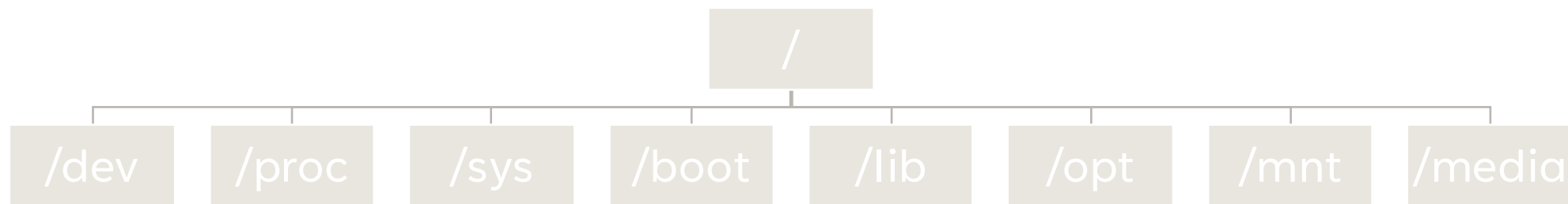
`rm -r dir` → recursive remove

`rm -rf dir` → force remove

LINUX FILE SYSTEM HIERARCHY



LINUX FILE SYSTEM HIERARCHY





FILE PERMISSIONS

Any permissions of a particular file is divided into three categories

Others permissions – People other than owner and group

Group permissions – People who are in the same group as owner/ people who are in the group listed for the file

User/Owner permissions – person who has created the file



FILE PERMISSIONS

There are three permission types

- `r` (read) - View file contents or list directory
- `w` (write) - Modify file or directory contents
- `x` (execute) - Run file as program or enter directory

Octal Notation is

$r = 4$ $w = 2$ $x = 1$

examples → 7 = all permissions, 5 = only read and execute



FILE PERMISSIONS

Command → `ls -l /etc/passwd`

```
-rw-r--r--. 1 root root 2847 Oct 15 10:23 /etc/passwd
```

```
|  |  |
```

```
|  |  └─ Others permissions (read)
```

```
|  └─── Group permissions (read)
```

```
└─── User/Owner permissions (read, write)
```




COMMANDS TO CHANGE FILE PERMISSIONS

There are two commands to change the file permissions

chmod – change mode

Command – to change the permissions of a specific file

chmod 777 filename or chmod u=rwx,g=rx,o=rx filename

chown – change owner (you have to be sudo)

Command – to change the owner/group owner of a specific file

chown user filename, chown user:group filename

HARD LINK VS SOFT LINK

Feature	hard link	Softlink
File Size	Same as original	Very small
Target Deleted?	File still accessible via hard link	Link becomes broken
Command	In source target	In -s source target

MASTERING SEARCHING THE FILES

There are 4 wild cards

*- Match any character

?- Match a single character

[]-try to match the characters within this bracket

{}- try to match words within this bracket

Examples for the first wild card *-

ls *.sh – any file that ends with .sh

ls s* - any file that begins with character s

Examples for the second wild card **?** –

ls f?n – any file that has f in the start and n in the end

ls ??? – any file that is exactly three characters long

Examples for the third wild card **[]** –

ls p[io]rn – list files with pi or po

ls f[og]r?e – list files with fo or fg

Examples for the fourth wild card **{}** –

ls {find,data}.txt – list files with either find or data

MASTERING GREP

Word vs Pattern :

Word example = the
pattern example = the, there, then,
therefore, thenguthe

Grep searches for patterns

Command = `grep "xxx"(pattern) filename`

Arguments that can be passed to search efficiently

- n = show the line number where you discovered the pattern
- c = show the count of lines
- i = ignore the case while searching
- v = inverse, show me the lines where the pattern is not present
- e = search for multiple patterns

MASTERING GREP WITH REGULAR EXPRESSIONS

Regular Expressions

- * - match 0 or n number of occurrences → `fil*e` = `fil`,`file`,`fi`,`fillle`
- . - match any one character → `fi.e` = `fire`, `file`, `fine`
- [] - match the characters inside brackets → `fi[ln]e` = `file`, `file`
- [^] - match all except the characters inside brackets
- ^ - match the lines that begin with → `^[0-9]` = any line beginning with 0-9
- \$ - match the lines that end with → `$sh` = `kush`,`aish`

Grep with -E argument

- + - match 1 or more time → `fil+e` = `file`,`fil`,`fillle`
- ? - match 0 or 1 time → `fil?e` = `fi`, `file`
- {m,n} - minimum value to maximum value → `fil{,3}e` = `fi`,`file`,`fil`,`fillle`

PIPING AND REDIRECTION

To combine commands together we use | this is called piping
examples

ls -l → is a command

grep web → is also a command

so → `ls -l | grep web` results in searching for a file called web

→ `ps aux | grep httpd` → filter httpd process

→ `cat file.txt | wc -l` → gives the number of lines

→ `netstat -tupln | grep LISTEN` → filtering LISTENING ports

→ `ls /usr/bin | wc -l` → gives the count

REDIRECTION

There are three types of redirection

0 – stdin – Standard input

1 – stdout – Standard output → usually this is displayed on screen

2 – stderr – Standard error → usually this is displayed on screen

To redirect you have to use ‘>’

examples

`man grep > ~/Desktop/grep.txt` → this is to write stuff, but if there is already stuff in that file, it will be overwritten

`man 2 read >> ~/Desktop/grep.txt` → using ‘>>’ is to append to already existing data

FINDING FILES MADE EASY WITH FIND

find Command is used to find files across the operating system

command → find path -name "filename"

arguments that can be used

find path -iname(case insensitive search) "filename"

→ all the regular expressions like *,?,[],{ },+ can be used here

find path -type f (files only)

find path -type d (directories only)

find path -perm /u+w (file/directory is writable by user)

find path -size +100M (larger than 100MB)

find path -size -1M (smaller than 1MB)

FINDING FILES MADE EASY WITH FIND

find Command is used to find files across the operating system

command → find path -name "filename"

find path -mtime -7 (modified less than 7 days ago)

→ similarly -ctime can be used for created less than 7 days ago

→ mmin, cmin can be used if the user wants to specify it in minutes

Execute command on the files found

find path -name "file" -exec command{} \;

{ } – takes each file and executes the command on it

\; - to exit the loop

USER AND GROUP MANAGEMENT

As a system admin you will be dealing with Users and Groups regularly.

Three types of users and 2 types of groups untayi

- ****root (UID 0):**** Superuser with unlimited privileges
- ****System users (UID 1-999):**** Run services and daemons
- ****Regular users (UID 1000+):**** Human users
- ****Primary group:**** Default group for user
- ****Supplementary groups:**** Additional groups for access

USER AND GROUP MANAGEMENT

Important files –

Info about users – /etc/passwd

Cat /etc/passwd

username:x(password):UID:GID:Comment:HomeDir:Shell

Command – useradd command/arguments

Arugments –

-g → Group

-G → subgroups(supplementary groups)

-u → UserID

-S → login shell (/sbin/nologin – frequently used in exam)

NOTE – Useradd command will not set the password. You have to use command → passwd

USER AND GROUP MANAGEMENT

To check which group the user belongs to →

id username

Modify existing user

Command → usermod command

Arguments →

-G → supplementary group

-aG → add supplementary group (a here means append)

Delete existing user

Command → userdel

Note → if you use userdel /home/user stays

Arguments →

-r → delete the user along with their home directory

PASSWORD INFORMATION

File → /etc/shadow

cat /etc/shadow → need superuser permission

username:\$hash\$:LastChange:Min:Max:Warn:Inactive:Expire

Password hash types:

\$1\$ - MD5 (deprecated)

\$5\$ - SHA-256

\$6\$ - SHA-512 (default)

To add a passwd to a newly created user

Sudo passwd username (if you want it to set it yourself use passwd)

Lock the account → usermod -l username

Unlock the account → usermod -u username

Force user to change the password on next login → usermod -e username

GROUP MANAGEMENT

Group Information:

```
cat /etc/group
```

```
groupname:x:GID:members
```

Create group

```
sudo groupadd -g xxx groupname
```

-g → groupID

Deleting groups:

```
Sudo groupdel groupname
```

Note → Cannot delete if it's primary group for any user

Adding user to multiple groups:

```
sudo usermod -aG group1,group2,group3 username
```

SIMPLE SUPERUSER AND SUDO

Superuser → anyone with admin privileges

Sudo → superuser do (execute this command with admin privileges)

How to add a user to sudo list?

→ You have to have root access in order to do this

Command → **sudo visudo** --> this lets you edit /etc/sudoers file

At the very end you can paste the command

Username ALL=(ALL:ALL) ALL

TRICKS

Remove a user from all the groups

(remember `-G` will overwrite not append so)

```
sudo usermod -G "" username
```

Force delete a user (even if logged in)

```
sudo userdel -f username
```

Creating multiple users at once (loop)

```
for user in user1 user2 user3; do
```

```
    sudo useradd -m -s /bin/bash -user
```

```
Done
```

PS – similar can be done for deleting multiple users

SYSTEM PROCESSES

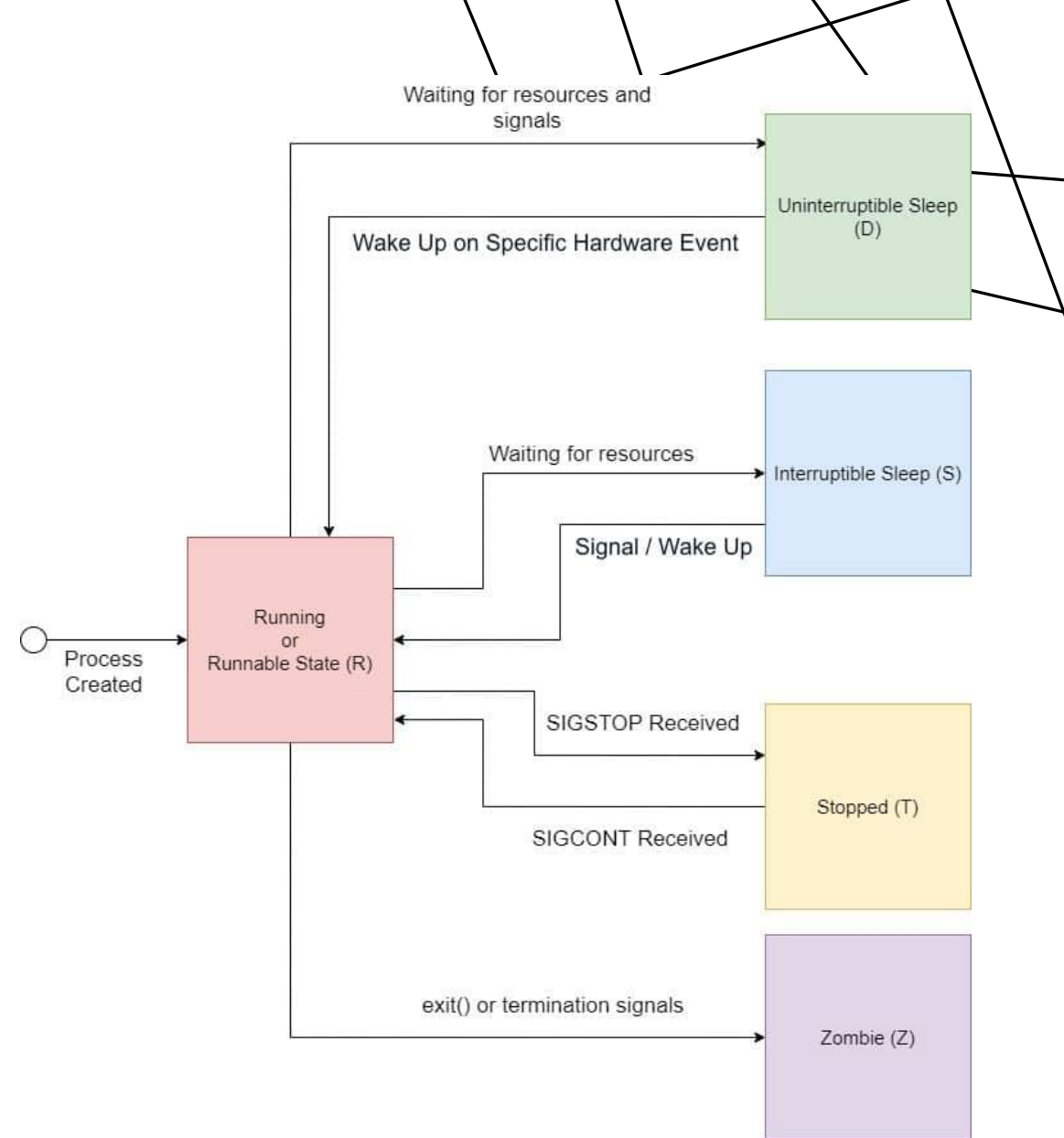
Process ante enti?

- A running instance of a program
- Each process has a unique PID (Process ID)
- Parent and child processes (process tree)
- Process states: running, sleeping, stopped, zombie

Why is it important to us?

- Monitoring and managing processes is vital for system health
- Controlling resource usage
- Killing unresponsive programs

Every PID (process ID) has a PPID(Parent Process ID)



PS COMMAND

Most common used command → `ps aux` or `ps lax`

The commands `aux` means

`a` = show processes for all users

`u` = show them in user-oriented style

`x` = show processes not attached to a terminal (daemons, services)

`l` = long format (add priority, nice value, flags etc)

There are two types of processes:-

Program Processes – shown in the foreground on the bash. example – `sleep 10`

Background processes – which run in the background. use command and then `'&'`

MORE ABOUT PROCESSES

To get all the ps in the background

Use command – Jobs

To bring a process from background to foreground –

Use command – fg %(xx) where xx is the number from jobs command

To send a process to background press ctrl + z

Note – doing so will stop the process

To resume a process in the background

Use command – bg %(xx) where xx is again the jobs number.

To kill the process(es) there are three commands

kill, killall, pkill

KILL, KILLALL, PKILL

All three commands can be used to kill a process

Examples – kill PID

killall sleep

kill processname

More commands

To forcefully kill a PS

Use command – kill -9 PID

or

killall -9 psname

or

kill -f PID

PROCESS MONITORING

There are two commands commonly used

Uptime & Top

uptime – Just an overview.

top – this is realtime data

Commands inside top:

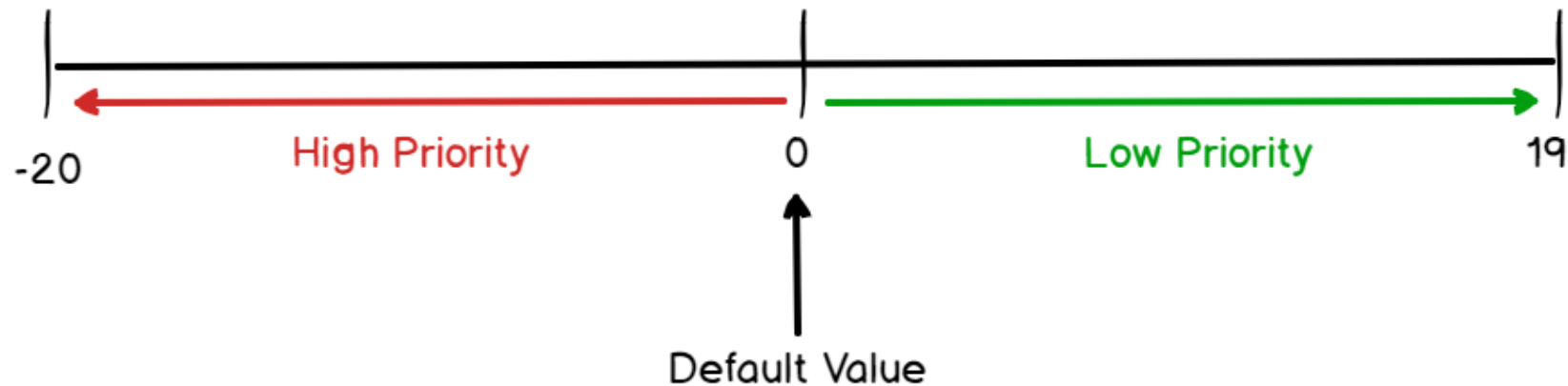
- `Shift + P` sort by CPU usage (windows – ctrl+)
- `Shift + M` sort by memory usage
- `k` kill process
- `q` quit

NICE & RENICE

Resources can be reallocated depending on their priority
i.e. Nice value

Scheduler → Linux → what process to do first

Niceness Scale on Linux



NICE & RENICE

Root user → 0 to -20

Regular user → 0 to 20

To give the command a nice value

Use command → `nice -n xx(value) command`

To adjust the priority of a running process

Use Command --> `renice -n xx() PID`

UMASK

with chmod and chown we can modify already existing files, but what if we want to change the default permissions for the future files

UMASK follows the inverse of numbers

1- execute 2- write 4- read

so umask 123 means users:read+write,
groups:read+execute, others:read

Note: for regular files, execute permission is not given, because those are not executable files by default.

STICKYBIT

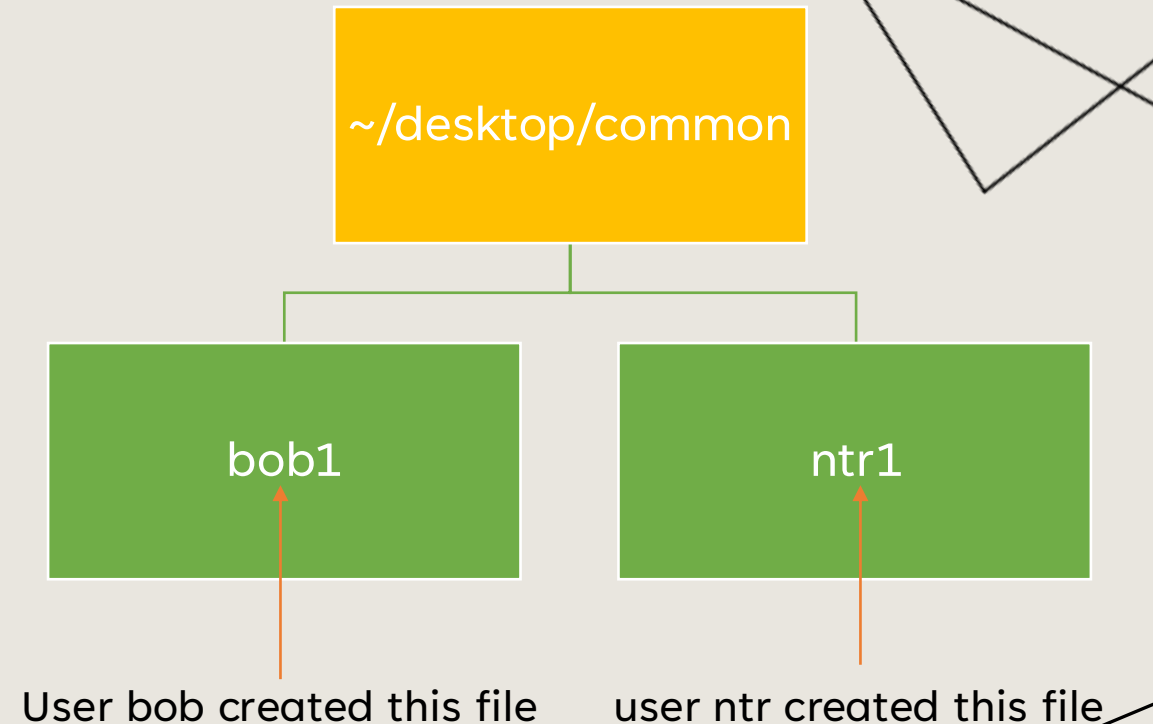
A special permission set on directories (not files) that restricts file deletion and renaming within that directory.

- Only the file's owner, the directory owner, or root can delete or rename files, even if others have "write" access.

Command – `sudo chmod 1xxx dirname` or `chmod +t dirname`

to check if sticky bit is applied use `ls -ld dirname`

Example – in the example right side, bob cannot delete what ntr has created and likewise the other way.



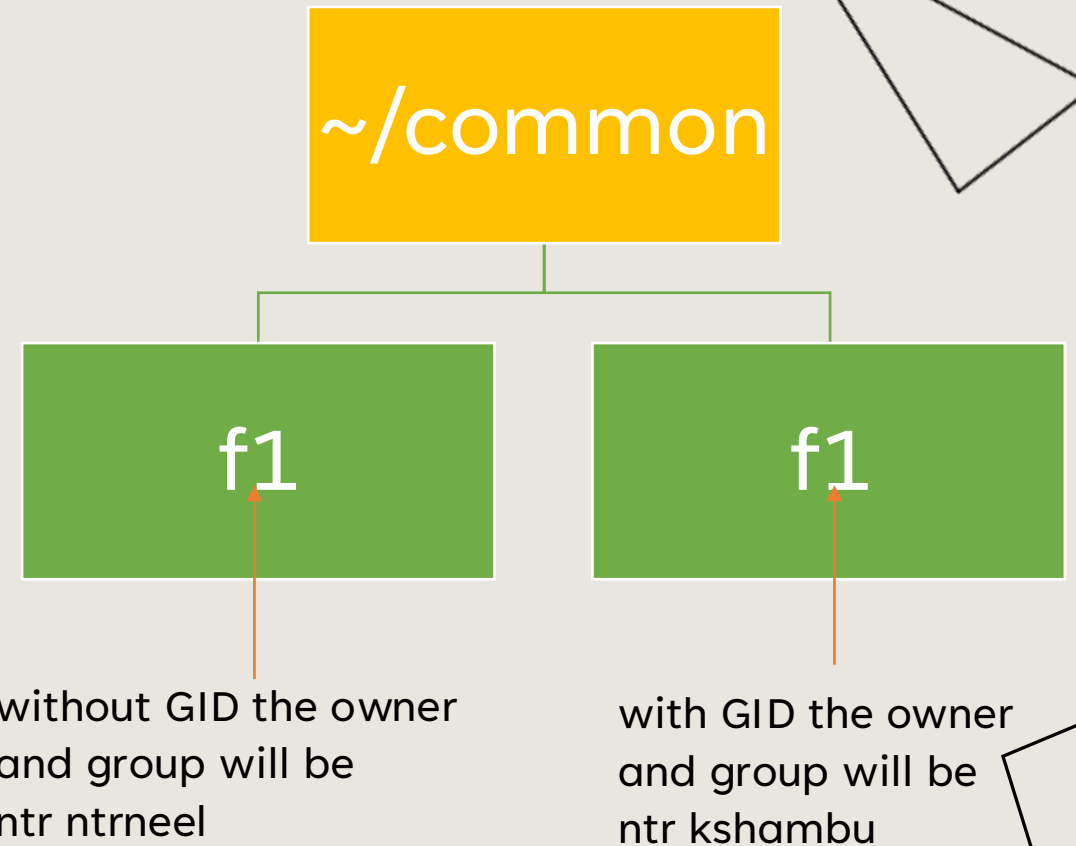
GID & SID

SetGID → but what does SGID do?

For example, if there is a directory belongs to the group ntrneel and the owner ntr, and someone else who was in the group spirit like example praboss, creates a file in the directory with SGID bit, the files are automatically belong to ntrneel group instead of spirit group.

Command → `sudo chmod g+s dirname`

how to check? --> `ls -l` shows 's'



GID & SID

SetSID → but what does SGID do?

This is when a person wants to execute a file/command with same permissions as owner.

example – changing the password of a user without sudo permissions.

→ if you go to `/usr/bin/passwd` you can see that command is already set to have SID bit because it wants every user on linux to run it as root so that they can change their password.

Command → `sudo chmod u+s filename`

CONFIGURING REPOS

Before you do anything else you have to register with redhat

Note – use the login credentials you used to login in redhat website

Commands – `sudo subscription-manager register`
→ `sudo subscription-manager status`

Now that you have registered the machine, you can begin downloading and installing repos.

Command – `sudo yum install reponame`

CONFIGURING REPOS - RHCSA

In the RHCSA exam you can't use yum install so you have to manually create a local repository

Steps to create a local repository and other things

1. create a local.repo file → `sudo vi /etc/yum.repos.d/local.repo`

2. inside that →

`name=(given in exam)`

`baseurl=xxx`

`gpgcheck=0` (this is usually used to verify GPG check)

`enabled=1` (1=enabled,0=disabled)

3.Clean and verify (optional)

`sudo yum clean all`

`sudo yum repolist`

MASTER YUM

very very easy

```
sudo yum install package_name
```

```
sudo yum search package_name
```

```
sudo yum list all | update / sudo yum update package_name
```

```
sudo yum remove package_name
```

ACL – ACCESS CONTROL LIST

Why ACLs?

ACLs allow fine-grained permissions for files and directories, beyond standard owner/group/other model.

→ To grant permissions to a specific user & a group.

→ em em ivvachu? r,w,x,X

→ now what is X? --> Give executable permissions for all the sub-directories inside the given directory but no execute permissions on the files. (given that the sub-directory already has executable permissions inherited from parent directory)

Command – `setfacl -m (modify) u:username:permissions filename`

How to check if it is applied.

`ls -l` → you should see a + sign, if you see this, all the user/ permissions are correct, but the group permissions are masked(not true)

ACL – ACCESS CONTROL LIST

To see what ACL is applied on the file

Command – `getfacl filename`

Recursive ACLs

Set the ACLs inside the directories as well.

Command – `setfacl R -m u:username:permissions(X) dirname`

Default ACL

With recursive the ACL is set to the subdirectories and files which were already created. Default is for the future files too

Command – `setfacl -m d:u:username:permissions(X) dirname`

Deleting ACL

If you want to remove ACL for a specific user – `setfacl -x u:username filename`

If you want to remove all the ACL applied – `setfacl -b dir/filename`

SYSTEMCTL

What is a DAEMON?

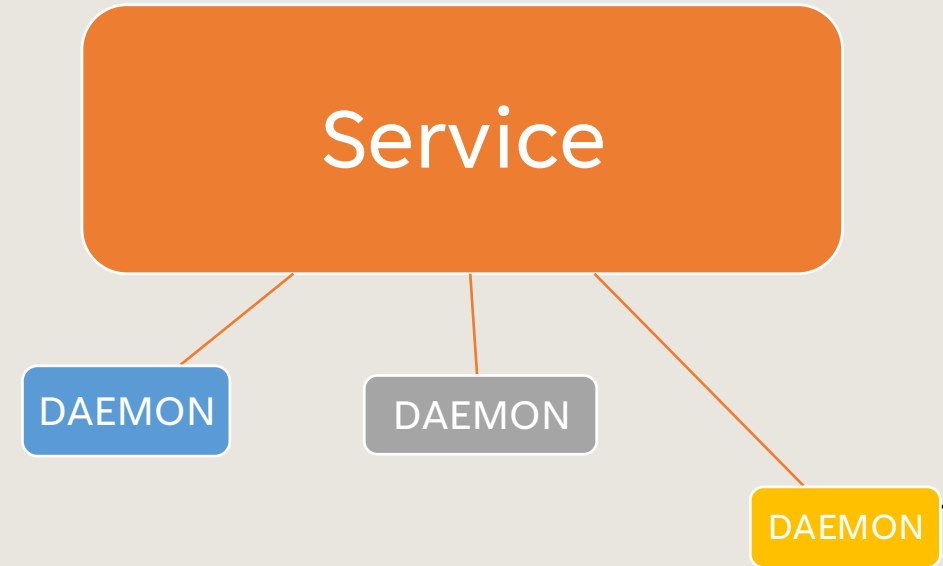
It is something that runs in the background, enabled at start time, usually ends with d

What is a Service??

Collection of one or more DAEMONS is a service

What is Systemctl?

It is a command to manage units.



SYSTEMCTL COMMANDS

STATUS	DESCRIPTION
ENABLED	boot time lo on chey bey!
DISABLED ACTIVE	boot time ledhu bokka ledhu ipudu nadustundhi
INACTIVE	SBI lunch time tarvatha raa!

Systemctl

status

start/stop/restart

enable/disable

is-active,is-enabled

NETWORK TIME PROTOCOL (NTP)

What is NTP?

NTP is a protocol used to synchronize the clocks of computers over a network. reason for correct time lol

Command – `timedatectl set-ntp true`

If you issue this command, linux enables automatic time synchronization, which is managed by `systemd-timesyncd`

If you want to manually change the time

Command – `timedatectl set-time "yyyy-mm-dd hh:mm:ss"`

before you issue this command do `timedatectl set-ntp false`

NETWORK TIME PROTOCOL (NTP)

If you want to manually change the timezone.

Steps to do it

1. issue the command – `tzselect` and select the correct timezone you want to change to
2. issue the command – `timedatectl set-timezone output(of the timezone)`

CHRONY

If your computer is not connected to the internet or your org has a NTP server and you want to use that instead of the default linux one. then follow the 4 steps.

1. `sudo yum install chrony`
2. `vi /etc/chrony.conf` – comment the line with pool and add a line `server <yourip> iburst`
3. `sudo systemctl restart chronyd`
4. `chronyc sources -v` to check the sources.

TUNING PROFILES

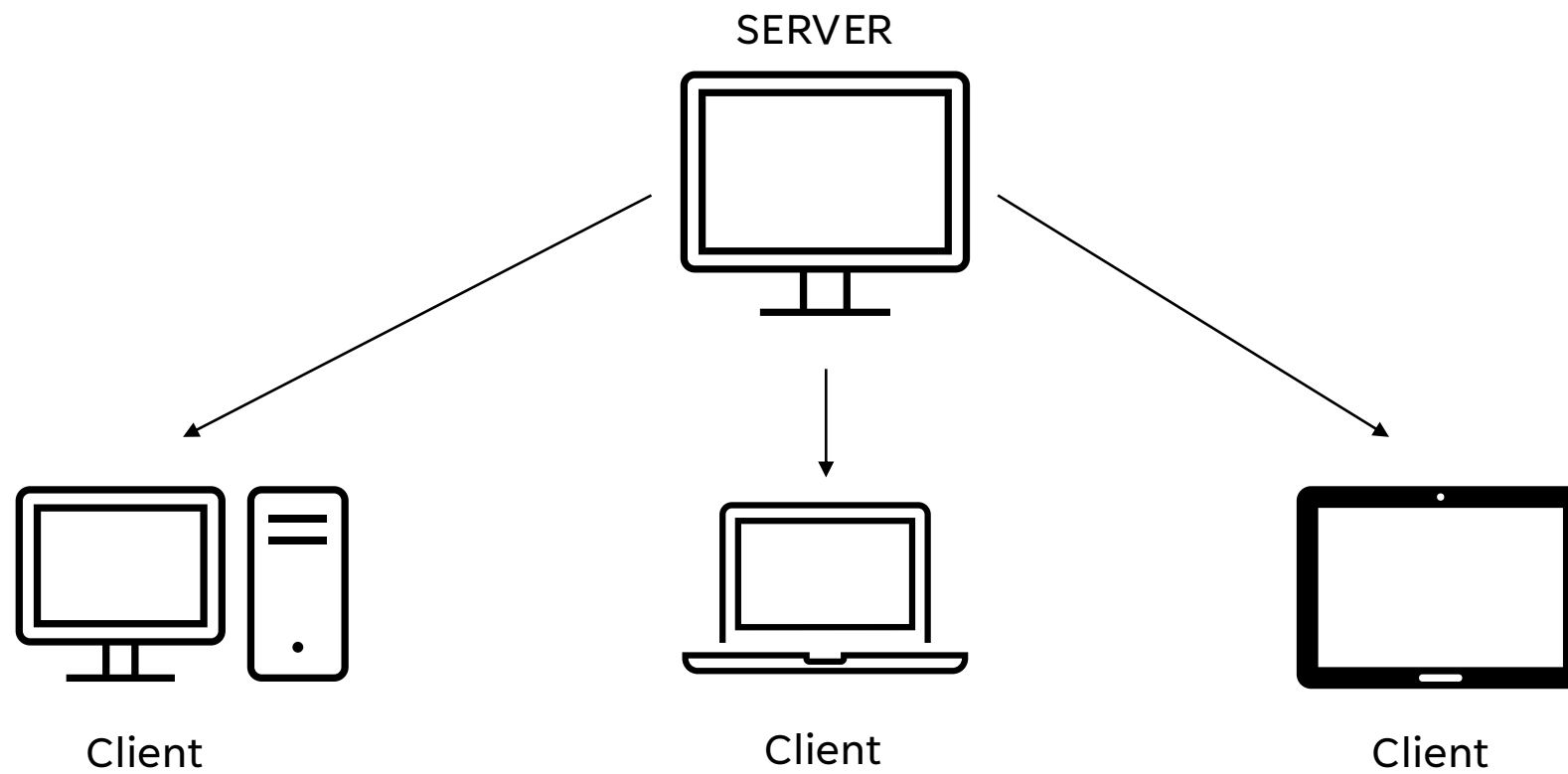
Same as game mode on your phone / battery saver mode on your gaming PC.

Command – tuned-adm profile xx

Note – if tuned daemon is not installed on your redhat, install it with `sudo yum install tuned` command and enable it using `systemctl enable tuned`

Tuning profiles commands →

command	description
list	lists all the profiles
active	show active profile
profile xx	activate the following profile
recommend	linux recommends you a profile based on the settings
off	turns off tuned daemon



WHY SSH?

SSH – secure shell protocol – remote login and command execution on servers.

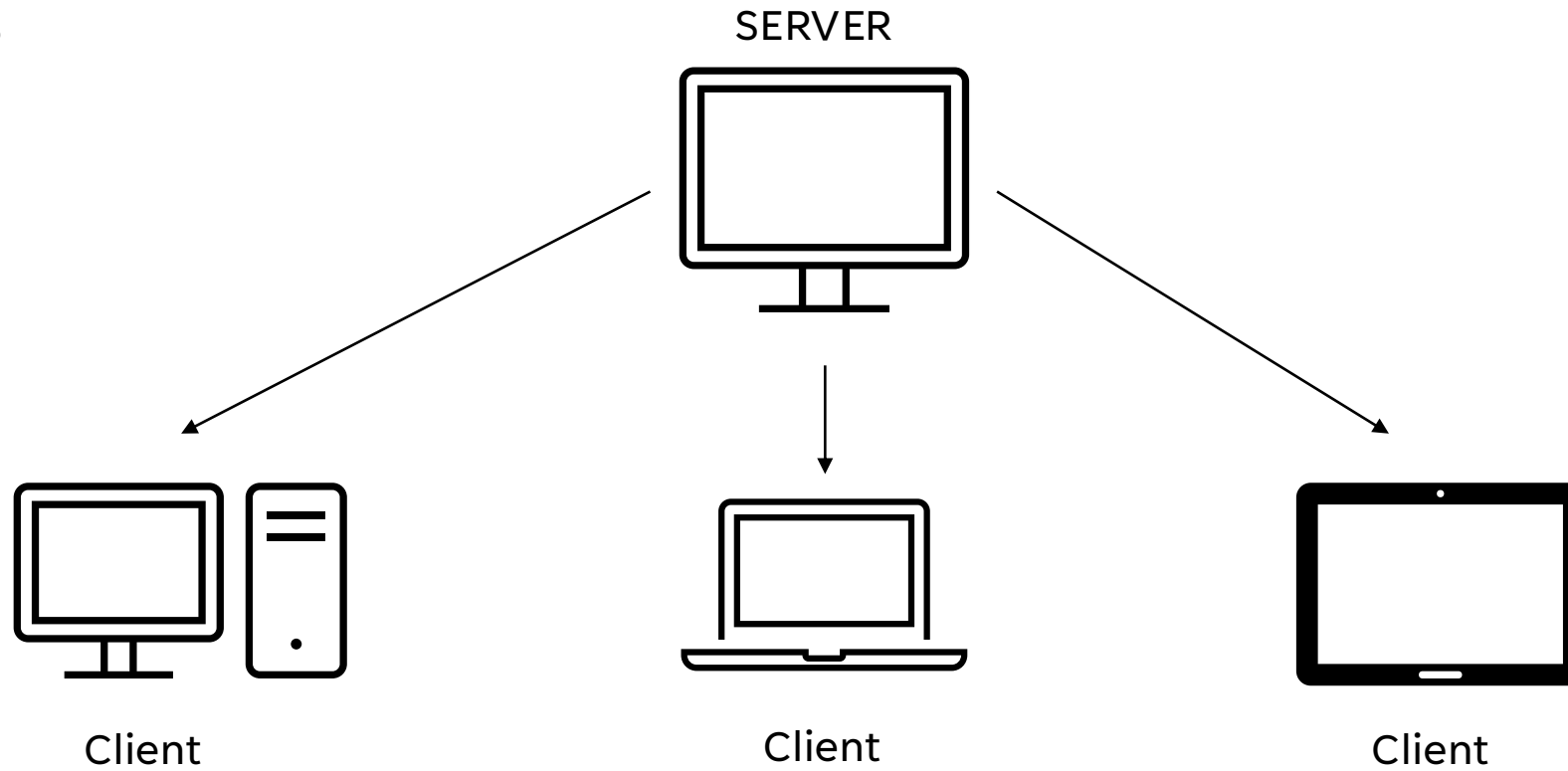
But why?

- it protects credentials and traffic from traffic and MITM (man in the middle) attacks
- It provides confidentiality(encryption), integrity and authentication.

Use case – Remote Login, SFTP, SCP, Port forwarding, automation

Command on the client machine – `ssh user@ipaddress`

Before we start today's
video. What is server?
what is client?
Terminology.



SCP – SECURE COPY PROTOCOL

Similar to SSH but just for transferring files one time.

Note: Files cannot be resumed if the connection resets while transferring.

Command – `scp filename user@ip:path -->` transferring a file from client to server

Vice versa – try transferring a file from server to client.

SFTP – SECURE FILE TRANSFER PROTOCOL

Similar to SSH but just for transferring files.

Note: Files can be resumed if the connection resets while transferring.

Command – `sftp user@ip -->` only to transfer files.

Commands inside sftp --

put	to transfer files from client to server
get	to transfer files from server to client

SELINUX

chmod, acl lu saripoinapudu idi osthadi mama. chala important topic jagrathaga vinu.

What is SELinux?

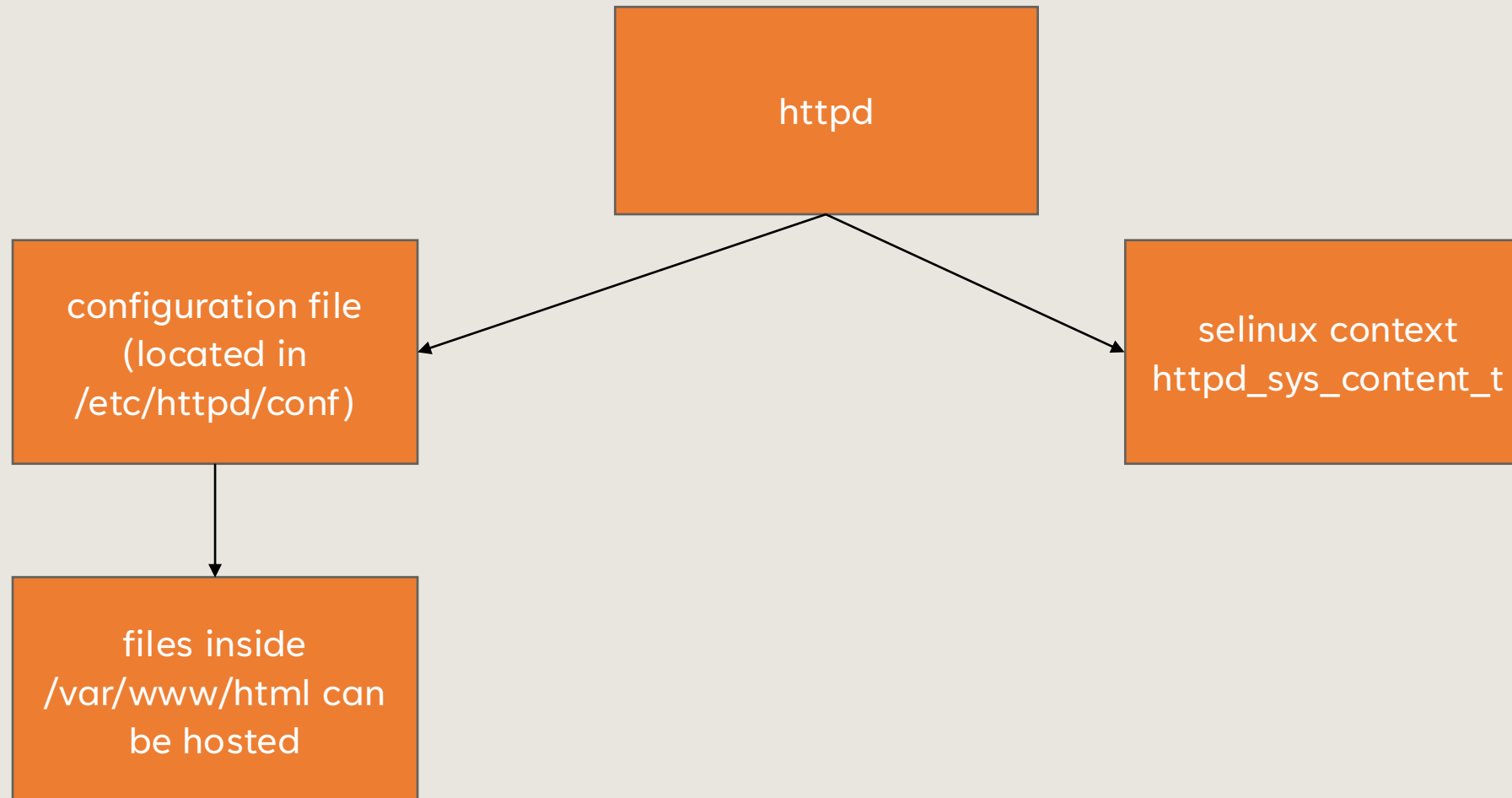
It is a collection of rules that cover which processes has access to which file, directory or port.

What is SELinux context?

It is the name used by SELinux policy to decide whether a process can access the file or no.

Let's look at an example.

EXAMPLE

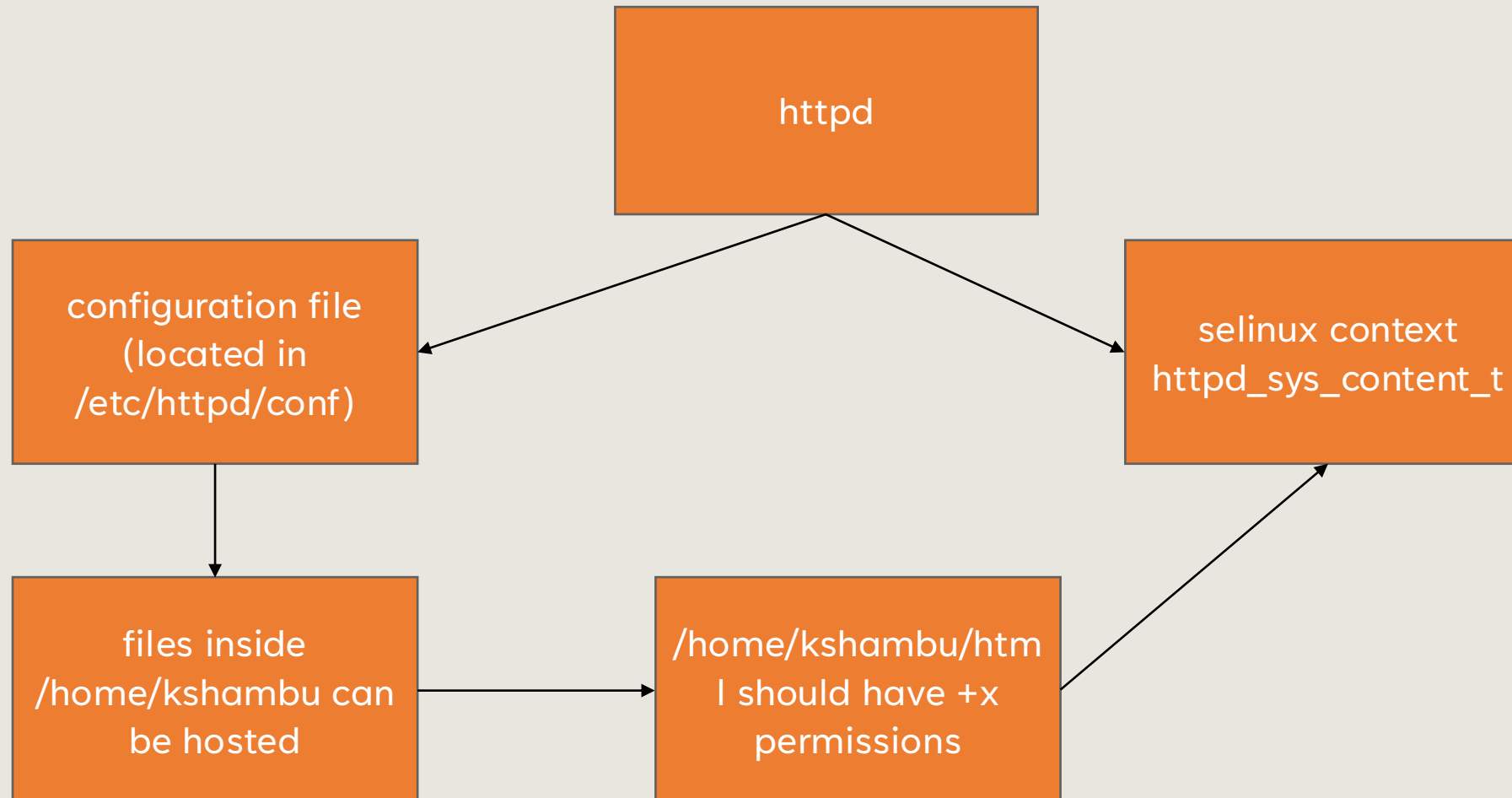


SELINUX

example -

1. is httpd daemon running? if not install it using yum, enable it and start it
2. if we check `/etc/httpd/config/httpd_config` where the config file is stored, we can see the apache server constantly searches for files in `/var/www/html` directory (DocumentRoot)
3. So, any file we create inside `/var/www/html` can be hosted.
4. Now what if we want to create a html file in our home directory? even if we change it in the `httpd_config` file(restart it after), it still won't allow us to host. Why is that?

EXAMPLE – CHANGE DIRECTORY



SELINUX

example -

1. this is because fcontext is not set to new directory that you have created to get access from httpd.
2. this can be checked **command** – ls -lZ for both new directory and /var/www/html

How to set the fcontext (SELinux context)?

Command – sudo semanage fcontext -a -t contextname "dir(.*)?"

restorecon -Rv dir

Note – other than / dir if you create it anywhere else, +x permissions should be given to root for SELinux to access that file.

SELINUX MODES

There are total of 3 modes for SELINUX

1. Enforcing → strict
2. Permissive → It's just logging what's going on
3. Disabled → is Disabled

Command to change mode Enforcing/Permissive –

setenforce permissive(xxx), getenforce will give you the current mode

To disable it → edit the file in /etc/sysconfig/selinux

Pro Tip – In the exam if you can't find where it would be use find

Command → find /etc -name "selinux"

SELINUX BOOLEAN

Ivi just toggle options laga on/off cheyadaniki.

One good thing – these are persistent across reboots

Command → `getsebool -a` will give you a list of all Booleans

Command → `setsebool xxx on/off` – temporary change

Command → `setsebool xxx -P on/off` – persistent change

DISK PARTITIONING

things we will be doing today

1. Creating a new partition
2. Installing a new file system
3. making the partition persistent
4. deleting the partition

assale **partition** ante endi? Indaka cheppa kadha, data organize cheyadaniki, multiple OS boot cheydaniki, swap space create cheyadaniki.

okay kushal **swap space** ante endi – RAM undha RAM? adi aipoinidi anuko ni ssd/hdd vadukuntadi computer.

DISK PARTITIONING

Steps to create a partition

1. List the disk available → `lsblk`
2. Select the disk → `fdisk /dev/sdc`(ni computer lo vere undochu)
3. Request new partition → click n(new) and p (primary)
4. Give a partition number → you can leave it default
5. Select first and last sector → first sector =default, last = size
note- size can be specified using +2K,M,G,T (kb, mb, gb, tb)
6. Give the partition type → linux (press t for more options)
7. Save the partition → press 'w'

DISK PARTITIONING

more steps

1. Initiate Kernel re-read part-table → `partprobe /dev/sdc`
2. make a file system for this partition → `mkfs.xfs(exfat and other file system read more on this) /dev/xxx`
3. mount the partition to a folder → `mkdir yyy, mount /dev/xxx xxx`
4. Make the mount persistent(fix chey)→`nano /etc/fstab`
add a new line → `/dev/xxx yyy xfs(file type) defaults 0 0`
5. check for errors → `mount -a`

DISK PARTITIONING

Deleting the partition

1. first chesina kalakandam reverse chey which is unmount the partition → `umount /dev/xxx`
2. next → `fdisk /dev/xxx` → press d
3. remove the additional line in the `/etc/fstab`
4. run `partprobe`

SWAP PARTITION

everything is same as making a disk partition

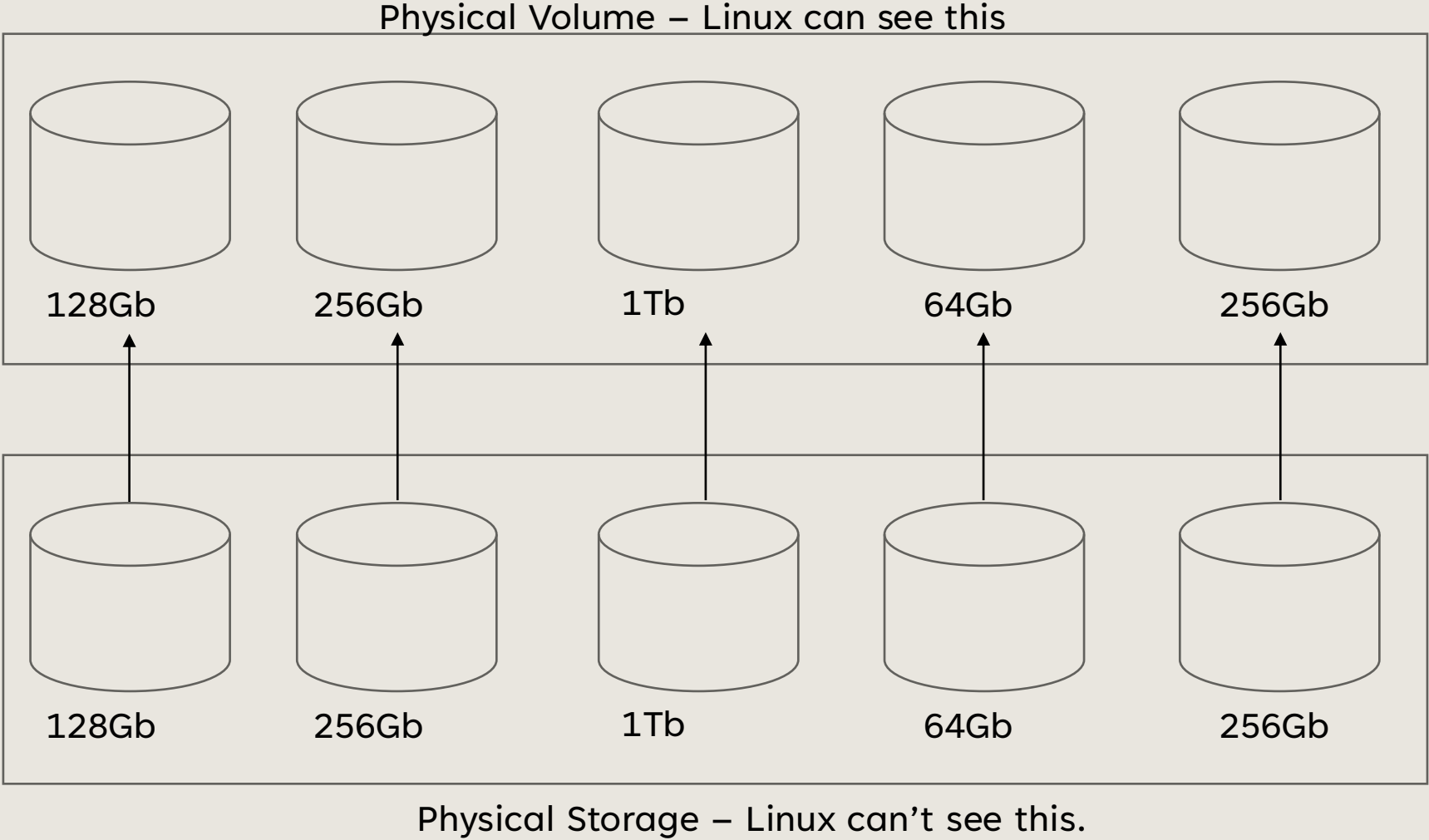
Only difference- after we select last sector instead of default linux you change to Linux Swap using 't' and press 'l' to get a list of all the options, select 82.

Another difference – instead of making a filesystem – mkfs, we do
mkswap /dev/xxx

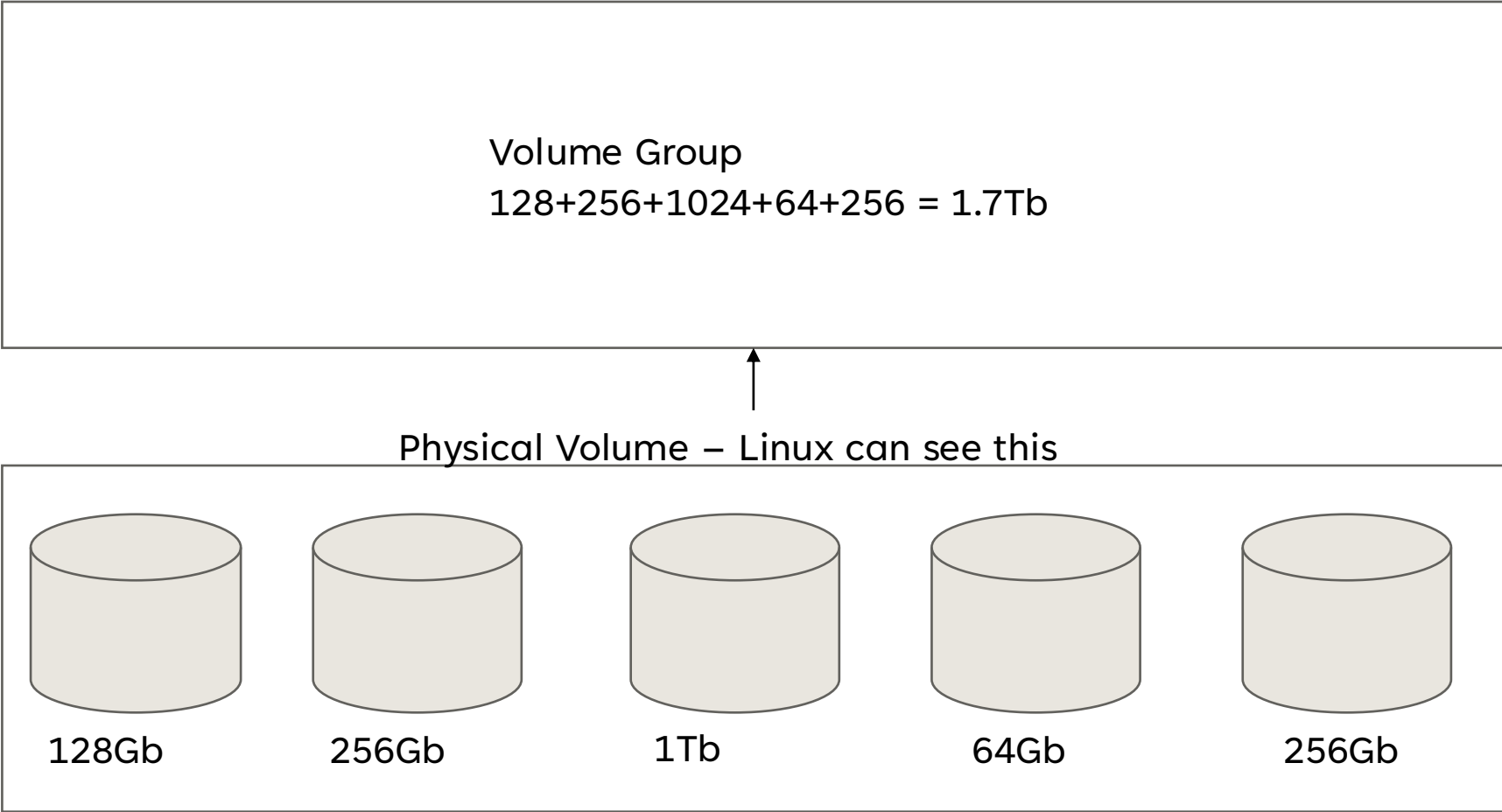
Another difference – editing /etc/fstab → we do
/dev/xxx swap swap defaults 0 0

Another difference – instead of mount, we do swapon -a

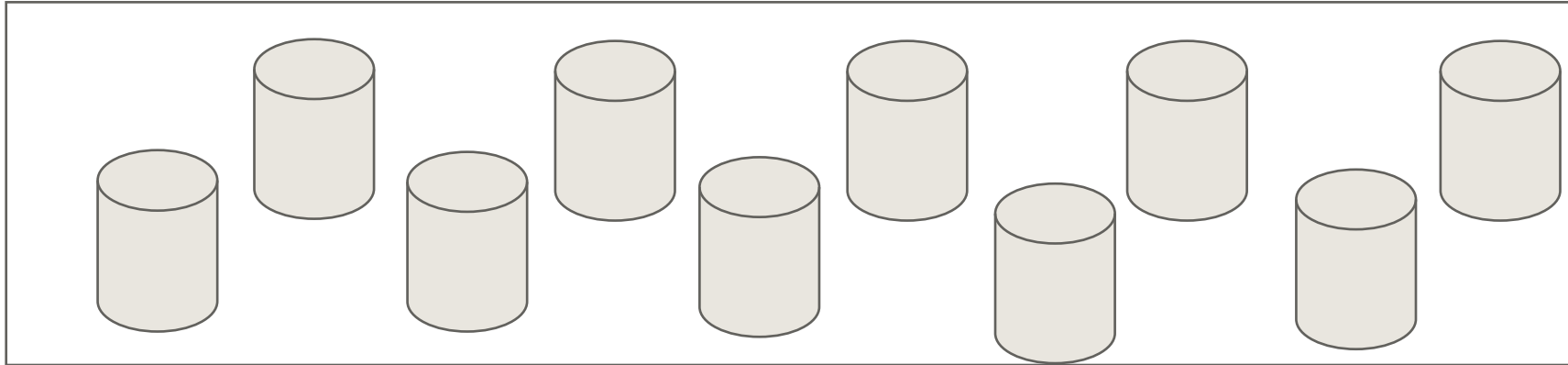
LOGICAL VOLUME



LOGICAL VOLUME



LOGICAL VOLUMES



Volume Group

$128+256+1024+64+256 = 1.7\text{Tb}$

LOGICAL VOLUME

process chudham oka sari, naa kosam okate sari.

first unna storage nundi physical volume cheyali,
etla cheyali? pvcreate /dev/xxx, pvcreate /dev/yyy

super ipudu create chesindi chudali ante → pvs

okate dani midha focus cheyali ante → pvdisplay /dev/xxx

next ochesi physical volume ni volume group ante gumpu lo
Govinda cheyali, etla? vgcreate vgname /dev/xxx /dev/yyy
/dev/zzz

again chesindi chudali ante → vgs

okate dani midha.... → vgdisplay

LOGICAL VOLUME

next ochesi volume group lo nundi manaki ishtam ochinantha storage tiskoni logical volume manaki nachinattu create cheskodame.

etla cheyali? `lvcreate -n name -L size vgroup`

super ipudu create chesindi chudali ante → `lv`

okate dani midha focus cheyali ante → `lvdisplay /dev/vgroup/lvname`

EXTENDING LV

scenario #1

ni volume group lo space undi

very easy – command → `lvextend -L size /dev/vgname/lvname`

filesystem already exists → use `-r`

scenario #2

add necessary physical volume and extend volume group

command → `vgextend vgname /dev/newpv` then do

`lvextend -r -L size /dev/vgname/lvname`

PRACTICE QUESTIONS

1. Create a cron job that appends the current date to the `/var/log/messages` log file. Ensure that this job runs at 10 PM on weekend. Execute the command `/usr/local/bin/yearly backup` at exactly 12 noon on Feb 4th.
2. Write the message "Welcome" into the file `/root/msg`, every three minutes between 9AM and 11AM, on every Friday in June.
3. You need to start a process called `my_script.sh` with a nice value of 10 to ensure it runs with lower priority.
4. A process with PID 1234 is consuming high CPU resources. You want to decrease its priority by changing its nice value to 15.

PRACTICE QUESTIONS

1. You are the system administrator of a server. The `/shared` directory is used by multiple users to collaborate, but you need to ensure the following:
 - a) All users can create files in the directory.
 - b) Users cannot delete or modify files created by others in this directory.
2. You are tasked with configuring a script `/shared/backup.sh` that performs backups requiring root privileges. The script needs to be executable by regular users, but it should always run with root privileges regardless of who executes it.

PRACTICE QUESTIONS

You are managing a directory /project where multiple users collaborate. The following requirements need to be met:

1. The owner of the directory, admin, must have full access (read, write, execute).
2. The group, Developers, must have read and write permissions to the directory.
3. A specific user, John, should have read-only access to the directory without affecting the permissions of others.

PRACTICE QUESTIONS

1. Configure the repositories. <http://xyz.server.com/baseos>,
<http://xyz.server.com/appstream>
2. Download the "chrony" package and ensure that the "chronyd" service starts automatically at system boot.
3. Set the system profile to "balanced"
4. Configure the NTP server to run at time google.com

PRACTICE QUESTIONS

1. Copy a file `/home/user1/data.txt` from the local machine to the remote server `192.168.1.100` under `/tmp` using SCP. Use SFTP to verify if the file exists on the remote server.
2. Check the current SELinux mode and switch it to permissive mode.
3. Allow the SSH service to use the new port 2222 in SELinux.
4. Configure the SELinux Boolean value `'httpd_can_network_connect'` to be enabled and ensure that this setting is persistent.

PRACTICE QUESTIONS

1. Create a partition of size 3GB from the disk `/dev/sdh`.
2. Extend LV1 by 1GB.

NETWORK CONNECTIONS

ee course lo nenu networking explain cheyanu, dinikosam IT course chesanu please adi check cheyandi before studying today's topic

https://www.youtube.com/playlist?list=PL3pyD7dDJWR354TWb_A4IOe6hYu-DuKNC

NETWORK CONNECTIONS

hostname

→ ni system peru enti?

To change → hostnamectl set-hostname xxx

ip address → ip a

ping → ping xxx.xxx.xx.xx

NETWORK CONNECTIONS

Adding/Modifying network connections

Command → **nmtui**

parameters to remember for exam → connection name, interface name (en0p), autoconnect, ipv4 address/network mask, gateway address, dns address

Note → after you add a connection you have to activate the connection to work.

FIREWALL

firewalld is the daemon that handles firewall

→ indulopala niku pre-defined zones anevi untayi like listed below.

Zone Name	Description
Trusted	This zone accepts all the incoming traffic. You can use this zone to manage the traffic on a trusted network because it will not filter anything.
Home	This zone is designed for only the home network. It permits only selected incoming traffic and reject all.
Work	This zone designed for only the work (corporate) networking. It permits only selected incoming traffic and reject all.
Internal	This zone intended to design for the internal network. It permits only what is allowed and rejects all.
Public	This zone rejects all the incoming traffic, except what is granted. Using with the default zone, we can add any newly network interfaces on it. It is designed to use only the public places.
External	This zone designed for outgoing traffic forwarded with masquerading is enabled. Also, we can use this for NAT
Dmz	This zone designed to use the demilitarized zone with limited public access. It permits only selected incoming traffic and reject all.
Block	This zone designed to reject all incoming traffic with an ICMP-host-prohibited message is returned. It permits only outgoing traffic.
Drop	This zone designed to drop all incoming traffic with no notification like ICMP errors. It is purely used in high secure places.

FIREWALL

to change these zones or add services, or add sources, or add ports the command we need to use is

Command → `firewall-cmd xxx --permanent(optional) --zone=yyy`(if you make any changes)

xxx can be replaced with → `--get-default-zone`, `--set-default-zone`, `--get-active-zone`, `--add-source`, `--remove-source`, `--add-service`, `--add-port`

After you have done that, you need to use `firewall-cmd --reload` to take effect.

q#1 Configure the firewall to allow HTTP and HTTPS traffic permanently.

q#2 Set the default zone to internal and ensure it persists after a reboot.

q#3 A service is running on a non-standard port (e.g., TCP 82). Allow this port permanently.

SHELL SCRIPTING

chala ante chala ante chala easy ga explain chestha vinu,

I am doing this assuming you don't have any like any programming experience or basics.

jaibob.py ki jaibob aney file ki difference extensions, in linux you doesn't care about **extensions**. (it's a text-based OS)

next are **data-types**. other languages, nu integer istunava, string istunava, character istunava, float, Boolean, array, tuple itla mention cheyalsi osthadi.

Linux lo po rareyyy ani lit tiskochu (name=student(no spaces))

SHELL SCRIPTING

practicals start chedham,

oh inko vishayam, nu ni code/script rasaka, danini regular file nundi executable file ki convert cheyali. ela how how u say?

Command → `chmod u+x filename`

SHELL SCRIPTING

if you want to read user input to store it in a variable

guess what

Command → read variablename ra unga

one more thing to note

double quotes = print the variable stored inside

single quotes = pushpa etla chepthey atla print cheyalna!

SHELL SCRIPTING – DAY02

Local Variables – These are only within a code block or a function. You can actually load these variables into your working shell using Source

Environmental Variables – These changes the behaviour of shell and User Interface. Examples - \$USER, \$HOME, \$SHELL

Positional Parameters – These are the arguments passed to a script from CLI

Examples - ./xxxx arg1 arg2 arg3 these are saved in \$0,\$1,\$2,\$3
Usually, \$0 refers to filename

\$n = nth argument password

\$* = Value of each argument

\$# = give me the number(total) of arguments passed)

SHELL SCRIPTING – DAY03

if else fi statements

very simple mama, anni programming languages lagane Bash lo kuda untadi

Structure

if[_condition_]

ikkada “_” = mandatory space

then

statements

fi(opposite of if)

SHELL SCRIPTING – DAY03

Operators

- gt = greater than
- ge = greater than or equal to
- lt = less than
- le = less than or equal to
- ne = not equal to
- eq = equal to

SHELL SCRIPTING – DAY03

String Comparisons

= → two strings are equal

example → if [_\$userinput=_bob_]

!= → not equal to

-n = is not null

-z = is null

SHELL SCRIPTING – DAY03

File Operations

- f = is a regular file
- d = is a directory
- e = exists
- r = has read permissions
- w = has write perm
- x = has execute perms
- s = has non-zero size

SHELL SCRIPTING – DAY03

if then elif then else fi

if else kakunda inkonni conditions rayali ani ankunte

Stucture

```
if[_condition1_]
```

```
then
```

```
    statements
```

```
elif[_condition2_]
```

```
then
```

```
    statements
```

```
else
```

```
    statements
```

```
fi
```

SHELL SCRIPTING – DAY03

`|| &&`

`||` - ante or annatu, like condition 1 or condition 2 might be true
in other words, renditlo edi true unna sare execute the statements

example – `if[_condition1_] || [_condition2_]`

`&&` - and, rendu conditions true aithene execute cheyyi

example - `if[_condition1_] && [_condition2_]`

SHELL SCRIPTING – DAY04 – LOOPS

Loops

assal loops enduku? simple mama, okate task malli malli cheyalsi ostundi ani ante loops anedi vaadtham

1) first di “for” loop -

Syntax 1 →

for variable in values

do

statements

done

SHELL SCRIPTING – DAY04

“for” loop -

Syntax 2 →

```
for((initialization;condition;decrement/increment))
```

```
do
```

```
    statements
```

```
done
```

SHELL SCRIPTING – DAY04

Next di While loop

ante, while condition is true, execute chey ra unga ani

Syntax →

```
while_[_condition_]
do
```

```
    statements
```

```
done
```

example if you want to print 1 to 9, you do while [x -lt 10]

SHELL SCRIPTING – DAY04

Next di Until loop

ante, until condition is True, execute chey ra unga ani

Syntax →

```
until[_condition_]
```

```
do
```

```
    statements
```

```
done
```

example if you want to print 1 to 9, you do until [x –ge 10]

FINAL DAY – ONLY NEEDED FOR RCHSA EXAM TAKERS

flatpak

Last konni videos mundu YUM nerchkunam, this topic very very similar to that.

7-8 commands anedi nu master cheyali anthe. avi kuda similar

1. `sudo dnf install flatpak` → idi run chey to install
2. `flatpak remote-add --if-not-exists --user(only for user) flathub https://url`
3. `flatpak remote-add --if-not-exists --user(only for user) RHEL https://url`
4. `flatpak remotes`
5. `flatpak search flathub/rhel firefox(name given on paper)`
6. `flatpak install flathub org.mozilla.firefox`
7. `flatpak uninstall flathub org.mozilla.firefox`
8. `flatpak upgrade flathub org.mozilla.firefox`
9. `flatpak list --app`

TOPICS I DID NOT COVER – RHCSA RHEL10

AutoFS

Resetting Root Password

emo konni SELinux topics vere la unnayi anta, IDC, linux is linux.

If you want to learn containers let me know it is important for RHEL 9 exam, but is removed in RHEL 10. Either way

pakkakiellutammuduavvakavvalikaasepunenu